

FLUID MECHANICS AND HYDRAULIC MACHINES

II B.TECH II Semester: MECHANICALENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME16	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OVERVIEW: This course is introduced with the fundamentals of fluid mechanics such as fluid and fluid flow properties followed by the discussion on the behaviour of fluid at rest and in motion. Then, the basic principles of mass, momentum and energy are emphasized. The differential equations of continuity and momentum are developed and their applications are highlighted in flow measurements. At the end, the basics of turbo machinery are also covered to provide glimps on hydraulic machines which plays an important role in the conversion of hydraulic energy to mechanical energy and vice-versa.								
COURSE OUTCOMES: The course should enable the students to 1. Describe the basic principles of fluid mechanics. 2. Identify various types of flows and formulate their governing equations. 3. Analyze the losses in pipe flows with the concepts of flow through pipes. 4. Evaluate hydrodynamic forces of jet striking different vanes from various angles. 5. Design the working proportions of hydraulic turbines and pumps.								
UNIT-I	FLUID STATICS							
Fluid Statics: Units and Dimensions, Properties of fluids: Mass Density, Specific weight, Specific Volume, Specific gravity, Viscosity, Surface tension and Capillarity, Vapour pressure and Cavitation. Pressure, Buoyancy and Floatation: Pressure and its Measurements (Manometers). Buoyancy Forces, Buoyancy forces on submerged bodies. Floatation and stability of floating bodies.								
UNIT-II	FLUID KINEMATICS							
Fluid Kinematics: Methods to describe fluid motions, Types of fluid flow- steady and unsteady, uniform and non-uniform, laminar and turbulent, rotational and irrotational, One dimensional, Two dimensional and Three Dimensional. Discharge and Continuity Equation: Rate of Flow or Discharge. Stream line, path line, streak line and stream tube, basic principles of mass, momentum and energy, continuity equation for one, two and three dimensional flow, fluid acceleration.								
UNIT-III	FLUID DYNAMICS AND FLOW THROUGH PIPES							
Fluid Dynamics: Surface and body forces, equations of motion: Euler's and Bernoulli's equations, Dimensional analysis, Bernoulli's equation for incompressible fluids and its practical applications: Venturimeter, Orificemeter, Pitot tube, Momentum equation. Flow through Pipes: Elementary turbulent flow (Reynolds's experiment), flow through pipes: head loses in pipes (Major losses & Minor losses). Head loss due to friction: Darcy Weisbach equation, Total Energy Line (TEL), Hydraulic Gradient Line (HGL), Pipes in series and parallel; Basic concepts of boundary layer and its Separation.								
UNIT-IV	BASICS OF TURBO MACHINERY							

Basics of Turbo machinery: Introduction, classification, hydrodynamic force of jets on stationary and moving plates, flat, inclined and curved vanes. Workdone and Efficiency: Jet striking centrally and at tip, velocity diagrams, work done and efficiency, flow over radial vanes.	
UNIT-V	HYDRAULIC TURBINES AND PUMPS
Hydraulic Turbines: Classification of Turbines, Heads and efficiencies, Impulse and Reaction turbines, Pelton wheel, Francis turbine and Kaplan turbine: working proportions, work done, efficiencies, draft tube theory, cavitation, surge tank, water hammer. Hydraulic Pumps: Classification, centrifugal and reciprocating pumps: working, work done, heads and efficiencies.	
Text Books:	
1. Modi and Seth, Hydraulics, Fluid Mechanics and Hydraulic Machinery, Rajsons Publications, 20th Edition, 2013. 2. YunusCengel, John Cimbala, Fluid Mechanics, McGraw Hill Education. 3. S K Som, Gautam Biswas and Suman Chakraborty, Introduction to Fluid Mechanics and Fluid Machines, 3rd Edition, McGraw Hill Education.	
Reference Books:	
1. Rajput, Fluid Mechanics and Hydraulic Machines, S.Chand& Co, 6th Edition, 1998. 2. Banga, Sharma, Hydraulic Machines, Khanna Publishers, 6th Edition, 2001. 3. Dr. R. K. Bansal, A Text Book of Fluid Mechanics and Hydraulic Machines, Laxmi Publications (P) Ltd, 10th Edition, 2015.	